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Bone fusion device, apparatus and method

Abstract

A bodiless bone fusion method, apparatus and device for insertion between bones that are to be fused together and/or in place of one or more of the bones, such as, for example, the vertebrae of a spinal column. The bodiless bone fusion device comprises one or more extendable plates, one or more extending blocks in communication with the extendable plates, one or more positioning elements for adjusting the extendable plates by manipulating the extending blocks, and one or more support panels for holding the positioning elements and guiding the extendable plates. The plates are able to be advantageously positioned in the confined space between the vertebrae to help brace the device until the bone has fused.

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Background/Summary

RELATED APPLICATIONS (1) This Application is a continuation of co-pending U.S. patent application Ser. No. 17/872,305, filed on Jul. 25, 2022 and entitled “BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD,” which is a continuation of U.S. patent application Ser. No. 16/789,128, filed on Feb. 12, 2020 and entitled “BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD,” which is a continuation of U.S. patent application Ser. No. 16/119,809, filed on Aug. 31, 2018 and entitled “BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD,” which is a divisional of U.S. patent application Ser. No. 14/210,094, filed on Mar. 13, 2014 and entitled “BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD,” which claims priority under 35 U.S.C. 119 (e) of the U.S. Provisional Application Ser. No. 61/794,789, filed Mar. 15, 2013, and entitled BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD” and the U.S. Provisional Application Ser. No. 61/858,505, filed Jul. 25, 2013, and entitled BODILESS BONE FUSION DEVICE, APPARATUS AND METHOD,” all of which are hereby incorporated by reference.

FIELD OF THE INVENTION

(1) This invention relates generally to bone fusion devices. More specifically, the present invention relates to bodiless devices for fusing vertebrae of the spine or other bones.

BACKGROUND OF THE INVENTION

(2) The spinal column is made up of vertebrae stacked on top of one another. Between the vertebrae are discs which are gel-like cushions that act as shock-absorbers and keep the spine flexible. Injury, disease, or excessive pressure on the discs can cause degenerative disc disease or other disorders where the disc becomes thinner and allows the vertebrae to move closer together or become misaligned. Similarly, vertebrae are able to weaken due to impact or disease reducing their ability to properly distribute forces on the spine. As a result, nerves may become pinched, causing pain that radiates into other parts of the body, or instability of the vertebrae may ensue.

(3) One method for correcting disc and/or vertebrae-related disorders is to insert a fusion cage as a replacement for and/or in between the vertebrae to act as a structural replacement for the deteriorated disc and/or vertebrae. The fusion cage is typically a hollow metal device usually made of titanium. Once inserted, the fusion cage maintains the proper separation between the vertebrae to prevent nerves from being pinched and provides structural stability to the spine. Also, the inside of the cage is filled with bone graft material which eventually fuses permanently with the adjacent vertebrae into a single unit. However, it is difficult to retain this bone graft material in the cage and in the proper positions to stimulate bone growth.

(4) The use of fusion cages for fusion and stabilization of vertebrae in the spine is known in the prior art. U.S. Pat. No. 4,961,740 to Ray, et al. entitled, “V-Thread Fusion Cage and Method of Fusing a Bone Joint,” discloses a fusion cage with a threaded outer surface, where the crown of the

thread is sharp and cuts into the bone. Perforations are provided in valleys between adjacent turns of the thread. The cage can be screwed into a threaded bore provided in the bone structure at the surgical site and then packed with bone chips which promote fusion.

(5) U.S. Pat. No. 5,015,247 to Michelson entitled, "Threaded Spinal Implant," discloses a fusion implant comprising a cylindrical member having a series of threads on the exterior of the cylindrical member for engaging the vertebrae to maintain the implant in place and a plurality of openings in the cylindrical surface.

(6) U.S. Pat. No. 6,342,074 to Simpson entitled, "Anterior Lumbar Underbody Fusion Implant and Method For Fusing Adjacent Vertebrae," discloses a one-piece spinal fusion implant comprising a hollow body having an access passage for insertion of bone graft material into the intervertebral space after the implant has been affixed to adjacent vertebrae. The implant provides a pair of screw-receiving passages that are oppositely inclined relative to a central plane. In one embodiment, the screw-receiving passages enable the head of an orthopaedic screw to be retained entirely within the access passage.

(7) U.S. Pat. No. 5,885,287 to Bagby entitled, "Self-tapping Interbody Bone Implant," discloses a bone joining implant with a rigid, implantable base body having an outer surface with at least one bone bed engaging portion configured for engaging between a pair of bone bodies to be joined, wherein at least one spline is provided by the bone bed engaging portion, the spline being constructed and arranged to extend outwardly of the body and having an undercut portion.

(8) U.S. Pat. No. 6,582,467 to Teitelbaum et al. entitled, "Expandable Fusion Cage," discloses an expandable fusion cage where the surfaces of the cage have multiple portions cut out of the metal to form sharp barbs. As the cage is expanded, the sharp barbs protrude into the subcortical bone of the vertebrae to secure the cage in place. The cage is filled with bone or bone matrix material.

(9) U.S. Pat. No. 5,800,550 to Sertich entitled, "Interbody Fusion Cage," discloses a prosthetic device which includes an inert generally rectangularly shaped support body adapted to be seated on hard end plates of vertebrae. The support body has top and bottom faces. A first peg is movably mounted in a first aperture located in the support body, and the first aperture terminates at one of the top and bottom faces of the support body. Further, the first peg projects away from the one of the top and bottom faces and into an adjacent vertebra to secure the support body in place relative to the vertebra.

(10) U.S. Pat. No. 6,436,140 to Liu et al. entitled, "Expandable Interbody Fusion Cage and Method for Insertion," discloses an expandable hollow interbody fusion device, wherein the body is divided into a number of branches connected to one another at a fixed end and separated at an expandable end. The expandable cage may be inserted in its substantially cylindrical form and may be expanded by movement of an expansion member to establish lordosis of the spine. An expansion member interacts with the interior surfaces of the device to maintain the cage in the expanded condition and provide a large internal chamber for receiving bone in-growth material.

(11) These patents all disclose fusion cage devices that can be inserted between vertebrae of the spine in an invasive surgical procedure. Such an invasive surgical procedure requires a long recovery period.

SUMMARY OF THE INVENTION

(12) The present application is directed to a bodiless bone fusion method, apparatus and device for insertion between bones that are to be fused together and/or in place of one or more of the bones, such as, for example, the vertebrae of a spinal column. The bodiless bone fusion device comprises one or more extendable plates, one or more extending blocks in communication with the extendable plates, one or more positioning elements for adjusting the extendable plates by manipulating the extending blocks, and one or more support panels for holding the positioning elements and guiding the extendable plates. The bodiless bone fusion device is able to be inserted between or replace the vertebrae by using a minimally invasive procedure. After the device has been positioned between the vertebrae, and the positioning elements are able to be rotated to

position the plates. In particular, the plates are able to be positioned by rotating the positioning elements causing extending blocks to move and push outwards against the plates as the extending blocks approach the ends of the bodiless bone fusion device. In some embodiments, a single plate is extended. Thus, the plates are able to be advantageously positioned in the confined space between the vertebrae to help brace the device until the bone has fused.

(13) A first aspect is directed to a bodiless bone fusion device for insertion into a desired location. The bodiless bone fusion device comprises an extending mechanism including one or more extending blocks mechanically coupled with a positioning element such that rotation of the positioning element causes the blocks to move with respect to the positioning element and a pair of plates straddling the extending mechanism and mechanically coupled with the extending blocks such that when the extending blocks move with respect to the positioning element, the plates move along a path with respect to each other between a retracted position in which the plates are adjacent to each other to an extended position in which the plates are spread apart from each other, wherein the plates are sized such that at least a portion of the perimeter of the plates about the path align with the outermost perimeter of the device about the path. In some embodiments, the plates are sized such that the entirety of the perimeter of the plates about the path align with the outermost perimeter of the device about the path. In some embodiments, the device further comprises one or more biasing elements physically coupled with both of the plates and positioned such that the biasing elements apply a force resisting the movement of the plates from the retracted position to the extended position. In some embodiments, the biasing elements have a shape selected from the group consisting of a ring, a C-shape and a ring-shaped coil. In some embodiments, the extending blocks each comprise an angled surface between a left side and a right side, wherein the left sides of the blocks are aligned with a left face of the plates and the right sides of the blocks are aligned with a right face of the plates. In some embodiments, angled surface forms a continuous sheet between the left and right sides of the blocks in order to increase the surface area of the angled surface. In some embodiments, the device further comprises a locking mechanism coupled with the positioning element and configured to physically bias the rotational orientation of the positioning element into one of a plurality of positions. In some embodiments, the locking mechanism comprises one or more stoppers each having a bump and a dial having one or more dimples and coupled with the positioning element such that the dial rotates with the positioning element, wherein the bumps do not rotate with the dial and the stoppers are positioned adjacent to the dial such that, when aligned, one or more of the bumps spring into one or more of the dimples. In some embodiments, the device further comprises one or more support panels coupled with the locking mechanism and the extending mechanism, wherein each of the support panels are positioned within a panel aperture on each of the plates such that as the plates move between the retracted and the extended positions the plates slide up or down the panels via the panels apertures. In some embodiments, at least one of the support panels comprises a pair of grip tabs that protrude from the sides of the support panel into a pair of grip apertures formed by the plates when the plates are in the retracted position.

(14) A second aspect is directed to a method of implanting a bodiless bone fusion device into a desired location. The method comprises inserting the bodiless bone fusion device in the desired location, wherein the bodiless bone fusion device comprises an extending mechanism including one or more extending blocks mechanically coupled with a positioning element such that rotation of the positioning element causes the blocks to move with respect to the positioning element and a pair of plates straddling the extending mechanism and mechanically coupled with the extending blocks such that when the extending blocks move with respect to the positioning element, the plates move along a path with respect to each other between a retracted position in which the plates are adjacent to each other to an extended position in which the plates are spread apart from each other, wherein the plates are sized such that at least a portion of the perimeter of the plates about the path align with the outermost perimeter of the device about the path and moving the plates

between the retracted position and the extended position with the extending mechanism. In some embodiments, the plates are sized such that the entirety of the perimeter of the plates about the path align with the outermost perimeter of the device about the path. In some embodiments, the bodiless bone fusion device further comprises one or more biasing elements physically coupled with both of the plates and positioned such that the biasing elements apply a force resisting the movement of the plates from the retracted position to the extended position. In some embodiments, the biasing elements have a shape selected from the group consisting of a ring, a C-shape and a ring-shaped coil. In some embodiments, the extending blocks each comprise an angled surface between a left side and a right side, wherein the left sides of the blocks are aligned with a left face of the plates and the right sides of the blocks are aligned with a right face of the plates. In some embodiments, the angled surface forms a continuous sheet between the left and right sides of the blocks in order to increase the surface area of the angled surface. In some embodiments, the bodiless bone fusion device further comprises a locking mechanism coupled with the positioning element and configured to physically bias the rotational orientation of the positioning element into one of a plurality of positions. In some embodiments, the locking mechanism comprises one or more stoppers each having a bump and a dial having one or more dimples and coupled with the positioning element such that the dial rotates with the positioning element, wherein the bumps do not rotate with the dial and the stoppers are positioned adjacent to the dial such that, when aligned, one or more of the bumps spring into one or more of the dimples. In some embodiments, the bodiless bone fusion device further comprises one or more support panels coupled with the locking mechanism and the extending mechanism, wherein each of the support panels are positioned within a panel aperture on each of the plates such that as the plates move between the retracted and the extended positions the plates slide up or down the panels via the panels apertures. In some embodiments, at least one of the support panels comprises a pair of grip tabs that protrude from the sides of the support panel into a pair of grip apertures formed by the plates when the plates are in the retracted position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a retracted perspective view of a bodiless bone fusion device according to some embodiments.
- (2) FIG. 1B illustrates an extended perspective view of a bodiless bone fusion device according to some embodiments.
- (3) FIG. 2 illustrates a cross-sectional view of components of the bodiless bone fusion device according to some embodiments.
- (4) FIG. 3A illustrates a profile view of the bodiless bone fusion device with the plates retracted according to some embodiments.
- (5) FIG. 3B illustrates a profile view of the bodiless bone fusion device with the plates extended according to some embodiments.
- (6) FIG. 4 illustrates a bodiless bone fusion device having a position locking mechanism according to some embodiments.
- (7) FIG. 5 illustrates a flow chart of a method of using the bodiless bone fusion device according to some embodiments.
- (8) FIG. 6A illustrates a front view of the bodiless bone fusion device having a loop biasing element according to some embodiments.
- (9) FIG. 6B illustrates a front view of the bodiless bone fusion device having a C shape biasing element according to some embodiments.
- (10) FIG. 6C illustrates a front view of the bodiless bone fusion device having a garter spring

biasing element according to some embodiments.

(11) FIG. 7 illustrates a side up-close view of a positioning element and stopper according to some embodiments.

(12) FIG. 8 illustrates a close-up view of support panels having retention tips according to some embodiments.

(13) FIG. 9A illustrates a retracted perspective view of a bodiless bone fusion device having stretched extending blocks according to some embodiments.

(14) FIG. 9B illustrates an extended perspective view of a bodiless bone fusion device having stretched extending blocks according to some embodiments.

DETAILED DESCRIPTION

(15) In the following description, numerous details and alternatives are set forth for purpose of explanation. However, one of ordinary skill in the art will realize that the invention can be practiced without the use of these specific details. For instance, the figures and description below often refer to the vertebral bones of a spinal column. However, one of ordinary skill in the art will recognize that some embodiments of the invention are practiced for the fusion of other bones, including broken bones and/or joints. In other instances, well-known structures and devices are shown in block diagram form in order not to obscure the description of the invention with unnecessary detail.

(16) FIGS. 1A and 1B illustrate retracted and extended perspective views, respectively, of a bodiless bone fusion device **100** according to some embodiments. The bodiless bone fusion device **100** is able to be constructed from a high strength biocompatible material, such as titanium, which has the strength to withstand compressive and shear forces in the spine that are generated by a patient's body weight and daily movements. Alternatively, part of all of the bodiless bone fusion device **100** is able to be constructed from one or more of the group consisting of high strength biocompatible material or a polymer such as PEEK, PEKK, and other polymeric materials known to be biocompatible and having sufficient strength. In some embodiments, the materials used to construct the bodiless bone fusion device include using additives, such as carbon fibers for better performance of the materials under various circumstances. The base biocompatible material is often textured or coated with a porous material conducive to the growth of new bone cells on the bodiless bone fusion device **100**.

(17) The bodiless bone fusion device **100** is able to have several conduits or holes **120** which permit the bone graft material to be inserted into the device **100** and to contact the vertebral bone before or after the device **100** has been inserted between the vertebrae of the patient. In particular, one or more holes **120** are able to be positioned on the lateral faces of the device **100** through one or both of the plates **102** such that the bone graft material is able to be inserted into the open spaces within the device **100** when the device is in the contracted position. It is understood that although only one conduit **120** on a lateral face is shown in FIG. 1A, any number of conduits **120** on lateral faces or other parts of the device **100** is contemplated. The bone graft material and the surface texturing of the device **100** encourage the growth and fusion of bone from the neighboring vertebrae. The fusion and healing process will result in the bodiless bone fusion device **100** aiding in the bridging of the bone between the two adjacent vertebral bodies of the spine which eventually fuse together during the healing period.

(18) As shown in FIGS. 1A and 1B, the bodiless bone fusion device **100** comprises one or more extendable plates **102**, one or more support panels **104**, one or more extending blocks **106**, one or more positioning elements **108** and one or more biasing elements **110**. The positioning element **108** is rotatably positioned within panel apertures **103** of the support panels **104** and operably coupled with the one or more extending blocks **106**. The support panels **104** are slidably positioned within plate apertures **118** of the extendable plates **102** and within a grip channel **114** of the extendable plates **102** when the device **100** is in the retracted position as shown in FIG. 1A. The biasing element **110** is positioned within biasing channels **112** on one or both ends of the extendable plates

102. In some embodiments, one or more of the holes **120**, the grip channels **114**, the biasing elements **110** and/or biasing channels **112** are able to be omitted. In some embodiments, one or more additional components are able to be added as are well known in the art. Additionally, it is noted that although FIGS. **1A** and **1B** only show two plates **102**, a single positioning element **108**, two extending blocks **106**, two support panels **104** and two biasing elements **110**, any number of plates **102**, positioning elements **108**, extending blocks **106**, support panels **104** and/or biasing elements **110** is contemplated.

(19) The one or more extending blocks **106** each are able to comprise a threaded conduit **122** for operably coupling to the positioning elements **108**. In particular, as described below, the positioning elements **108** are able to comprise a plurality of threaded screws having different diameters wherein the threaded conduits **122** of the extending blocks **106** are able to be configured to screw onto or otherwise engage with one of the threaded screws of the positioning elements **108**. Alternatively, one or more of the screws are able to have the same diameter. Further, each of the extending blocks **106** are able to comprise angled upper and/or lower outer surfaces for contacting/engaging angled inner surfaces **123** (see FIGS. **3A** and **3B**) of the extending plates **102**. Specifically, the angled outer surfaces are able to be configured such that as the blocks **106** move along the positioning element **108** the angles outer surfaces push against the angled inner surfaces **123** causing the plates **102** to move outwards.

(20) The support panels **104** are able to be sized/configured to slidably fit within one or more plate apertures **118** within the extendable plates **102**. In some embodiments, one or more of the plate apertures **118** extend completely through the corresponding plate **102**. Alternatively, one or more of the plate aperture **118** are able to only extend partially through the corresponding plate **102**. When in the retracted position, the top and bottom portions of the support panels **104** are able to be positioned fully within a plate aperture **118** of each of the extendable plates **102** (e.g. such that the edge of the support panels **104** is substantially flush with the surface of the plates **102** if the plate aperture **118** extends through the top of the plate **102**). As the plates **102** are extended outward to the extended position, the plates **102** slide up the panels **104**, but the panels **104** remain at least partially within the plate apertures **118** even when in the fully extended position. In some embodiments, as shown in FIG. **8**, the top and/or bottom of the panels **104** comprise one or more retention tips **101** that bow out or otherwise protrude out from the top and/or bottom of the panels **104** in order to block or mechanically stop the plates **102** from sliding off the top of the panels **104**. For example, the retention tips **101** are able to extend out from the panels **104** and if the plates **102** slide up to the retention tips on the panel **104**, the tips **101** provide a biasing force that pushes the plates **102** back down the panels **104** until they no longer contact the retention tips **101**.

Alternatively, other types of fasteners or stopping mechanisms are able to be used to prevent the plates **102** from sliding of the panels **104** as are well known in the art.

(21) As a result, the panels **104** are able to maintain the alignment of the plates **102** with each other and with the positioning element **108** and extending blocks **106**. Also, as described above, the support panels **104** are each able to comprise one of the panel apertures **103** such that the panels **104** are able to receive one end of the positioning element **108**. Specifically, the panel apertures **103** are able to be configured to receive a non-threaded portion of an end of the positioning element **108** such that the positioning element **108** is held in place relative to the support panels **104**, but allowed to rotate within the panel apertures **103**. One or more of the support panels **104** are also able to comprise one or more grip tabs **105** that extend out the sides of the support panels **105**. As described below, the grip tabs **105** are configured to fit within the grip channels **114** of the plates **102** and provide a gripping point to an insertion instrument used to insert and otherwise manipulate the device **100**. In some embodiments, the grip tabs **105** comprise one or more indentations, conduits and/or fasteners for receiving detachably coupling with an insertion tool. For example, the grip tabs **105** are able to be configured such that they create a profile that matches the profile of the insertion tool such that the tool is able to securely grip the device **100** via the grip tabs **105**.

(22) The extendable plates **102** are able to be located on opposite sides of the device **100** and face in opposite directions. Internally, the plates **102** are able to have one or more angled inner surfaces **123** (see FIGS. 3A and 3B) that have end thicknesses that are larger than their middle thicknesses such that the thickness of the angled surfaces **123** gradually increases while going from the middle to the ends of the plate **102**. Alternatively, the angled inner surfaces **123** are able to be configured such that they have end thicknesses that are smaller than their middle thicknesses such that the thickness of the angled surfaces **123** gradually decreases while going from the middle to the ends of the plate **102**. In either configuration, the angled surfaces **123** are able to interact with the extending blocks **106** to cause the plates **102** to retract or extend between the retracted and extended positions. As described above, the plates **102** each comprise one or more plate apertures **118** that are sized to slidably receive the top or bottom of the support panels **104**. As a result, the panels **104** are able to keep the plates **102** in alignment with each other as the plate **102** slide up and down along the support panels **104**. Additionally, in some embodiments the panels **104** are able to be shaped similar to the grip tabs **105** and/or other shapes such that the panels **104** are able to both support the plates **102** as well as enable the plates **102** to slide along the panels **104**.

(23) As also described above, the plates **102** each able to comprise the one or more biasing channels **112**. In particular, the biasing channels **112** are able to be configured such that when the device **100** is in the retracted position the biasing channels **112** of the plates **102** align to form a continuous channel that crosses between the plates **102**. In some embodiments, the biasing channels **112** are able to align at two or more positions between the plates **102** to form a continuous loop or other shape that crosses multiple times between the plates **102**. In some embodiments, the biasing channels **112** include a lip guard **111** that holds the biasing elements **110** within the biasing channels **112**. Alternatively, the biasing channels **112** are able to comprise coupling elements (not shown) that enable the biasing elements **110** to directly couple to the biasing channels **112** in order to stay within the channels **112**. Although as shown in FIG. 2 the lip guard **111** is substantially straight forming a square-like channel **112**, it is contemplated that the guard **111** is able to be angled, rounded, indented or otherwise shaped such that the guard **111** is able to retain the biasing elements **110** within the biasing channels **112**. Further, the biasing channels **112** are able to each include one or more portions that are nonparallel to the direction in which the plates **102** are able to be extended in order to fit a biasing element **110** that provides resistance to the extension of and biases the plates **102** in the retracted position. In some embodiments, as shown in FIGS. 1A and 1B the biasing channels **112** form a C shape. Alternatively, the biasing channels **112** are able to form a loop (see FIGS. 6A-6C), snake or other shapes having nonparallel portions as are well known in the art. Alternatively, the biasing channels **112** are able to be entirely parallel but be coupled to the biasing element **110** such that a nonparallel portion is unnecessary to provide the force resisting extension of the plates **102**. In some embodiments, the biasing channels **112** are positioned on the ends of the plates **102** as shown in FIGS. 1A and 1B. Alternatively, one or more of the biasing channels **112** are able to be positioned on another lateral face or faces of the plates **102**.

(24) Additionally, the plates **102** are able to have serrated edges or teeth **136** to further increase the bodiless bone fusion device's gripping ability and therefore ability to be secured in place between the bones for both a long-term purchase and a short-term purchase. In some embodiments, the serrated edges or teeth **136** are able to be in a triangular or form a triangular wave formation as shown in FIG. 2. Alternatively, the serrated edges or teeth are able to be filleted, chamfered, or comprise other teeth shapes or edge waves as are well known in the art. As described above, the plates **102** are able to comprise the grip channels **114** positioned on opposite sides of one or more ends of the plates **102**. The grip channels **114** are able to be configured such that when the device **100** is in the retracted position the grip channels **114** of the plates **102** align and are partially filled by grip tabs **105** of the support panels **105**. The remainder of the grip channels **114** is able to be configured to receive gripping fingers of an insertion instrument (not shown). In particular, the grip channels **114** enable the insertion instrument to grip the grip tabs **105** of one of the support panels

104 to manipulate the device **100** and to prevent the device **100** from slipping or during insertion into a patient. Alternatively, the grip tabs **105** are able to comprise one or more screw holes or other types of fasteners for fastening to an insertion instrument as are well known in the art.

(25) Finally, the plates **102** are able to be configured such that when in the retracted position the extendable plates **102** house or surround the remainder of the components of the device **100**. As a result, the bodiless bone fusion device **100** provides the advantage of maximizing the plate size to device size ratio because the size of the plates **102** is equal to the size of the device **100** in the retracted position creating a 1 to 1 ratio. This enables the device **100** to incorporate larger plates **102** that increase stability and surface area, which would not be possible with devices that incorporate a body. Additionally, it should be noted that one or more of the plates **102** are able to be non-flat, non-parallel to each other, or otherwise non-uniform. For example, one or more of the plates **102** are able to be partially or fully concave, convex and/or angled. Further, in some embodiments one or more of the plates **102** are able to be adjustable or interchangeable such that they enable adjustments to their surface/body shape.

(26) The positioning element **108** is able to comprise a positioning aperture **109**, a first screw **107A** and a second screw **107B** coupled together (see FIG. 2). The positioning aperture **109** is configured to receive a drive/engaging mechanism of a tool (not shown) such that the tool is able to rotate the positioning element **108**. The positioning aperture **109** is able to comprise numerous shapes and sizes as are well known in the art. Alternatively, the positioning aperture **109** is able to be omitted and/or the end of the positioning element **108** is able to be shaped to fit within the drive/engaging mechanism of the tool. The first screw **107A** is threaded opposite of the second screw **107B**. For example, if the first screw **107A** is left threaded, the second screw **107B** is right threaded or vice versa. Furthermore, the first screw **107A** is of a slightly different size than the second screw **107B**. As described above, the positioning element **108** is able to be operably coupled to one or more of the extending blocks **106**. For example, a first one of the extending blocks **106** is able to be threaded onto the first screw **107A** and a second one of the extending blocks **106** is able to be threaded on to the second screw **107B**.

(27) When coupled to the positioning element **108**, the extending blocks **106** are able to be positioned in the middle of the bodiless bone fusion device **100** in the retracted position. When the positioning element **108** is turned appropriately, the extending blocks **106** each travel outwardly on their respective screws **107A** and **107B**. As the extending blocks **106** travel outwardly, they push the angles surfaces **123** of the plates **102** causing the plates **102** to extend outward along the support panels **104**. In other words, the inner plate surface **123** when in contact with the extending blocks **106** act in such a manner so as to push the respective plates **102** apart. Thus, the plates **102** will be fully extended when the extending blocks **106** reach the opposite ends of the screws **107A**, **107B**. To retract the plates **102**, the positioning device **108** is turned in the opposite direction and the extending blocks **106** will each travel back to the middle on their respective screws **107A** and **107B**. It is contemplated that the operation of the device **100** is able to be reversed such that the plates **102**, extending blocks **106**, and positioning element **108** are configured such that the extending blocks **106** travel inwardly to extend the plates **102** into the extended position and travel outwardly to retract the plates **102** into the compact position. In any case, the nonextended plates **102** of the bodiless bone fusion device **100** provide a compact assembly that is suitable for insertion into the patient's body through a open, or minimally invasive surgical procedure. As used herein, an open or a minimally invasive procedure comprises a procedure wherein a smaller surgical incision is employed as compared to the size of the incision required for conventional invasive surgery, for example arthroscopic procedures. Moreover, minimally invasive procedures minimize or eliminate the need for excessive retraction of a patient's tissues such as muscles and nerves, thereby minimizing trauma and injury to the muscles and nerves and further reducing the patient's recovery time.

(28) The biasing elements **110** are able to be configured to fit within the biasing channels **112** of

two or more plates **102** when the plates **102** are in alignment. For example, as shown in FIGS. **1A**, **1B** and **6B**, one or more of the biasing elements **110** are able to be shaped in a C shape or broken loop shape. Alternatively, as shown in FIG. **6A**, one or more of the biasing elements **110** are able to have a circular, oval or loop shape. Alternatively, as shown in FIG. **6C**, one or more of the biasing elements **110** are able to have a garter spring shape or any other type of shape formed by the biasing channels **112**. Further, the biasing element **110** are able to be shaped to fit behind the lip guard **111** such that the lip guard **111** holds the biasing element **110** in place within the biasing channels **112**. Alternatively, the biasing element **110** is able to directly couple to the plates **102** in order to stay within the biasing channels **112**. In some embodiments, the biasing elements **110** are able to be structured and/or positioned such that their body blocks the extension of the plates **102** and thus the extension of the plates **102** causes deformation and/or stretching of the body of the biasing elements **110**. As a result, the body deformation and/or stretching resistance of the biasing elements **110** provides an extension-resisting force that biases the plates **102** in the retracted position. This biasing provides the advantage of ensuring that the plates **102** remain in contact with extending blocks **106** as the plates **102** are extended and/or retracted. In some embodiments, one or more of the biasing elements **110** comprise nitinol to provide the deformation resistant and/or flexible structure. Alternatively, the biasing elements **110** are able to comprise other material having deformation resistant, springing and/or elastic properties as are well known in the art.

(29) FIG. **2** illustrates a cross-sectional view of components of the bodiless bone fusion device **100** according to some embodiments. As shown in FIG. **2** and described above, the positioning element **108** is able to comprise a first screw **107A** and a second screw **107B** wherein the first screw **107A** is threaded differently than that of the second screw **107B** and is a different size than the second screw **107B**. For example, in some embodiments the first screw **107A** is an 8-32 screw and the second screw is a 6-32 screw. A first extending block **106A** and a second extending block **106B** are utilized with the positioning element **108** to extend and retract one or more of the plates **102** with respect to each other and/or the positioning element **108**. The first extending block **106A** has an internal opening and threading to fit around the first screw **107A**. The second extending block **106B** has an internal opening and threading to fit around the second screw **107B**. The support panels **104** are coupled with the positioning element **108** via the plate apertures **118** of the plates **102**. Specifically, because the plate apertures **118** receive the ends of the support panels **104**, they prevent the panel apertures **103** of the support panels **104** from moving axially with respect to the positioning element **108** thereby keeping the ends of the positioning element **108** within the panel apertures **103**. Further, the plates **102** are each coupled with each other via the support panels **104** that maintain the alignment of the plates **102** and the biasing elements **110** that hold the plates **102** onto the support panels **104**.

(30) FIG. **3A** illustrates a profile view of the bodiless bone fusion device **100** with the plates **102** retracted according to some embodiments. When the extending blocks **106** are positioned in the middle of the positioning element **108** with the first screw **107A** and the second screw **107B**, the plates **102** are positioned adjacent and/or in contact with each other. FIG. **3B** illustrates a profile view of the bodiless bone fusion device **100** with the plates **102** extended according to some embodiments. As shown in FIG. **3A**, the bodiless bone fusion device **100** is compressed/retracted when the extending blocks **106** are in the middle of the bodiless bone fusion device **100**. As a user rotates the positioning element **108** via the positioning aperture **109**, the extending blocks **106** gradually move outward from the middle. If the user turns the positioning element **108** in the opposite direction, the extending blocks move back towards the middle. As the extending blocks **106** are moving outward, the extending blocks **106A**, **106B** push on inner angled surfaces **123** of the plates **102**. The plates **102** extend because the extending blocks **106** exert force against the angled inner surfaces **123** of the plates **102** outwardly as shown by the arrows **140**. When the extending blocks **106** are positioned near the ends of the bodiless bone fusion device **100**, the plates **102** extend beyond the outer edges of the ends of the support panels **104** of the bodiless bone

fusion device **100** and ultimately secure the bodiless bone fusion device **100** between two bones. (31) In operation, the bodiless bone fusion device **100** is initially configured in a compact position such that the extending blocks **106A**, **106B** are located in the middle of the bodiless bone fusion device **100** thereby allowing the plates **102** to contact each other and/or the edges of the ends of the support panels **104** to be substantially flush with the outer surfaces of the plates **102** through the plate apertures **118**. The compact bodiless bone fusion device **100** is then inserted into position within the patient and surgeon is able to expand the bodiless bone fusion device **100** by rotating the positioning element **108** which moves the extending blocks **106A**, **106B** towards the opposing ends of the bodiless bone fusion device **100**—one near the head of the positioning element **108** and the other towards the tail of the positioning element **108**. As the extending blocks **106A**, **106B** move away from the middle, the plates **102** are pushed outwardly from the pressure of the extending blocks **106A**, **106B** against the angled inner surfaces **123**. Eventually the extending blocks **106A**, **106B** exert a satisfactory force between the extended plates **102** and the bones to be fused. At that point the bodiless bone fusion device **100** is able to remain in place. If the plates **102** are extended too far, the surgeon is able to rotate the positioning element **108** in the opposite direction moving the extending blocks **106A**, **106B** back towards the middle. At the same time, the biasing elements **110** exert a retraction force in the opposite direction of the force **140** that ensures the plates **102** retract as the extending blocks **106A**, **106B** move back towards the middle of the device **100**. In particular, the retraction force is able to be applied to the plates **102** by biasing elements **110** throughout operation of the device **100** in order to both keep the plates **102** from sliding off the support panels **104** and keep the plates **102** in contact with the extending blocks **106** as the blocks **106** move along the positioning element **108**. Thereafter, material for fusing the bones together is inserted through the holes and openings **120** within the bodiless bone fusion device **100**.

Alternatively, the insertion of the material for fusing the bones together is able to be omitted.

(32) FIG. 4 illustrates a bodiless bone fusion device **400** having a position locking mechanism **402** according to some embodiments. The bodiless bone fusion device **400** shown in FIG. 4 is substantially similar to the bodiless bone fusion device **100** except for the differences described herein. It is noted that the plates **102** of the bone fusion device **400** have been omitted from FIG. 4 for the sake of clarity. As shown in FIG. 4, at least one of the support panels **104** comprises one or more additional panel apertures **99** configured to receive a position locking mechanism **402**, wherein the position locking mechanism **402** comprises one or more dials **404** and one or more stoppers **406**. The dial **404** is configured to rotatably fit within the panel apertures **99** and comprises a dial aperture **412** and one or more dimples **410** along the edge or perimeter of the dial **202**. The dial aperture **412** is able to be sized or otherwise configured to receive an end of the positioning element **108** such that if the positioning element **108** is within the dial aperture **412**, the end of the positioning element **108** will cause the dial **404** to rotate along with the positioning element **108**. In some embodiments, the positioning element **108** causes the dial **404** to rotate by directly physically contacting the dial aperture **412**. Alternatively, the positioning element **108** is able to cause the dial **404** to rotate via indirect contact. The one or more dimples **410** are able to be configured to receive one or more bumps **408** of the stoppers **406**. In particular, the dimples **410** are able to have concave dimensions that substantially match convex dimensions of the bumps **408**. The stoppers **406** are able to be configured to fit within the panel apertures **99** adjacent to the dial **404** and comprise one or more bumps **408**. The stoppers **406**, dials **404** and apertures **99** are configured such that when within the apertures **99**, the stoppers **406** are adjacent or in contact with the dial **404** and the bumps **408** of the stoppers **406** snap or spring fit within the dimples **410** of the dial **404** when a dimple **410** and a bump **408** are aligned. Additionally, when a dimple **410** and a bump **408** are not aligned, the bump **408** is compressed against the dimple-less edge of the dial **404** and primed to spring or decompress into a dimple **410** when alignment is achieved.

(33) In some embodiments, the dial **404** is held in place within the additional panel apertures **99** by force applied by the bumps **408** of the stoppers **406**. For example, in some embodiments the

dimples **410** are able to be concave and centered along the perimeter of the dial **404** such that when the bumps **408** are within the dimples **410** the outer walls of the concavity of the dimples **410** prevents the dial **404** and/or the stoppers **406** from falling out of place. As another example, as shown in FIG. 7 the dial **404** is able to be omitted or incorporated into the positioning element **108**, wherein the perimeter of the positioning element **108** that is adjacent the stoppers **406** forms a trough or channel **401** that receives the stoppers **406** such that the positioning element **108** is unable to come out of position with respect to the stoppers **406**. In such embodiments, the bottom of the trough is able to comprise the dimples **410** for receiving the bumps **408** of the stoppers **406**. Alternatively, the dial **404** is able to be otherwise coupled or uncoupled within the apertures **99** by one or more fastening elements as are well known in the art.

(34) In some embodiments, the stoppers **406** are held in place within the additional panel apertures **99** by place holders **407**. In particular, the place holders **407** are able to be tensioned and/or compressed by the wall of the apertures **99** when the stoppers **406** are inserted into the apertures **99** and thus provide a spring force against the walls of the apertures **99** to try and relieve that tensioning/compression. Accordingly, the spring force holds the stoppers **406** within the apertures **99**. Alternatively, one or more of the stoppers **406** are able to be otherwise coupled or uncoupled within the apertures **99** by one or more fastening elements as are well known in the art. Although as shown in FIG. 4, the device **400** comprises one of the panels **104** including the position locking mechanism **402**, wherein the position locking mechanism **402** comprises a single dial **404** having sixteen dimples **410** and two stoppers **406**, it is understood that any number of the panels **104** are able to include a position locking mechanism **402** and the position locking mechanism is able to include any number of dials **404** having any number of dimples **410** coupled to any number of stoppers **406**. In some embodiments, the additional panel apertures **99** are able to replace the panel aperture **103** and/or the dial aperture **410** is able to be substantially similar to the panel aperture **103** in size and shape.

(35) In operation, as the positioning element **108** is rotated to extend or retract the plates **102**, the dial **404** is rotated along with the positioning element **108** and the bumps **408** compress and decompress into and out of the dimples **410** as they move in an out of alignment with the bumps **408**. As a result, each point during the rotation of the positioning element **108** that results in an alignment of a bump **408** and a dimple **410** serves as a demarcated degree of rotation and/or degree of extension/retraction of the plates **102**. In this way, the position locking mechanism **402** provides the advantage of enabling a user to rotate the positioning element **108** and thereby extend the plates **102** to predetermined rotation/extension amounts and/or by predetermined rotation/extension intervals represented by the spacing and number of dimple **410** and bump **408** alignment points. For example, the position and/or number of dimples **410** and/or bumps **408** of the position locking mechanism **402** is able to be adjusted to adjust the number and/or position of the alignment points and therefore the number and/or position of plate extension points. Thus, the position locking mechanism **402** of the bodiless bone fusion device **400** is able to be tuned to different size devices **400** based on the number of extension increments needed and the desired extension distance interval between each of the increments. In some embodiments, the increments are configured to be constant. Alternatively, the increments are able to be configured to decrease in size as the plates **102** approach the maximum extension level. Alternatively, other increment profiles are able to be used as are well known in the art. Further, the compression of the bumps **408** and their resistance thereto during rotation of the positioning element **108** between alignment points provides a slipping resistance force that resists unintended rotation of the positioning element **108** out of an alignment point. As a result, the position locking mechanism **402** provides the advantage of reducing the chance of the positioning element **108** unintentionally rotating and/or the plates **102** unintentionally extending or retracting.

(36) FIG. 5 illustrates a flow chart of a method of using a bodiless bone fusion device according to some embodiments. A user pre-configures the one or more plates **102** of the bodiless bone fusion

device to the retracted position with the positioning element **108** and the one or more extending blocks **106** such that the device has a minimized form factor at the step **502**. The user inserts the bodiless bone fusion device into a desired position in between the bones at the step **504**. The user extends the plates **102** to a desired extension level between the bones by rotating the positioning element **108** causing the extending blocks **106** to push the plates **102** outward to the desired extension level at the step **506**. In some embodiments, the rotating of the positioning element **108** comprises rotating the positioning element **108** through a number of alignment points of the position locking mechanism **402** until a desired alignment point is reached. As a result, the method is able to provide the benefits of a minimally invasive surgery due to the minimized form factor of the bodiless bone fusion device in the retracted position and a more accurate and stable extension point due to the position locking mechanism.

(37) FIGS. **9A** and **9B** illustrate a retracted perspective view and an extended perspective view of a bodiless bone fusion device **900** having stretched or expanded extending blocks according to some embodiments. The bodiless bone fusion device **900** shown in FIGS. **9A** and **9B** is substantially similar to the bodiless bone fusion device **100** except for the differences described herein.

Specifically, the sides **902** of the extending blocks **106** shown in FIGS. **9A** and **9B** extend such that the sides **902** substantially align with the outer surface of the device **900**. As a result, the extending blocks **106** span the entire width of the plates **102** which creates greater surface area for the blocks **106** to contact the plates **102** as well as greater stability in the extended position as a wider portion of the plates **102** is directly contacted/supported by the blocks **106**. In such embodiments, the skirt or sides **904** of the plates **102** are able to comprise a block cavity **906** configured to receive the sides **902** of the blocks **106** when the device **900** is in the retracted position. Although as shown in FIGS. **9A** and **9B**, both sides **902** of both blocks **106** are expanded to align with the exterior surface of the sides **904** of the plates **102**, one or more of the sides **902** of one or more of the blocks **106** are able to not be expanded and/or be expanded less. For example, one of the sides **902** of one of the blocks **106** is able to extend part way into the cavity **906** on one of the sides **904** of the plates **102**.

(38) Thus, the bodiless bone fusion device, apparatus and method described herein has numerous advantages. Specifically, the bodiless bone fusion device provides the advantage of maximizing the plate size to device size ratio because the size of the plates is equal to the size of the device in the retracted position creating a 1 to 1 ratio. This enables the device to incorporate larger plates that increase stability and surface area, which would not be possible with devices that incorporate a body. Also, the device provides the advantage of the grip channels that ensure the non-slippage of the driving mechanism during the operation of the bone fusion apparatus. Further, the position locking mechanism provides the advantage of reducing the chance of the positioning element unintentionally rotating and/or the plates unintentionally extending or retracting. Also, as mentioned above, the method of use requires only a small incision and minimally invasive surgical procedure advantageously promoting health and rapid recovery by the patient. Indeed, bone growth occurs around the bodiless bone fusion device and particularly at the locations of the extended plates, such that the bodiless bone fusion device is further secured by the bone growth, which further promotes a superior, robust bone fusion result. Moreover, the device provides the advantage of extending blocks that span the entire width of the plates thereby creating greater surface area for the blocks to contact the plates as well as providing greater stability in the extended position as a wider portion of the plates is directly contacted/supported by the blocks.

(39) The present invention has been described in terms of specific embodiments incorporating details to facilitate the understanding of principles of construction and operation of the invention. Such reference herein to specific embodiments and details thereof is not intended to limit the scope of the claims appended hereto. It will be apparent to those skilled in the art that modification may be made in the embodiments chosen for illustration without departing from the spirit and scope of the invention.

(40) For example, it should be noted that although the above bodiless bone fusion devices are

described in reference to a pair of extending blocks, a pair of screws, and wherein each plate is shaped such that the ends are larger than the middle, and the size of the plate gradually increases while going from the middle to the ends, the use of a single extending block in the above embodiments is contemplated. Specifically, if using a single extending block, the above embodiments would operate the same except the positioning element would comprise a single screw that when engaged would cause the single extending block to move from one end of the screw to the other end thereby exerting a force against the plates such that they move into the extended position. In such embodiments, each plate is shaped such that one end is larger than the opposite end, and the size of the plate gradually increases going from the smaller end to the larger end.

Claims

1. A bone fusion device for insertion into a desired location comprising: an extending mechanism including a plurality of extending blocks coupled with a positioning element such that movement of the positioning element causes the plurality of extending blocks to move with respect to the positioning element; a first plate; and a second plate, the first and second plates coupled with the plurality of extending blocks such that when the plurality of extending blocks move with respect to the positioning element, the first and second plates move along a path with respect to each other between a retracted position in which the first and second plates are adjacent to each other to an extended position in which the first and second plates are spread apart from each other, wherein a portion of the first plate directly contacted by the plurality of extending blocks changes as the plurality of extending blocks move with respect to the positioning element.
2. The device of claim 1, wherein the first and second plates are sized such that an entirety of a perimeter of the first and second plates about the path align with an outermost perimeter of the device about the path.
3. The device of claim 1, further comprising one or more biasing elements physically coupled with both of the first and second plates and positioned such that the one or more biasing elements apply a force resisting the movement of the first and second plates from the retracted position to the extended position.
4. The device of claim 3, wherein the one or more biasing elements have a shape selected from the group consisting of a ring, a C-shape and a ring-shaped coil.
5. The device of claim 1, wherein the plurality of extending blocks each comprise an angled surface between a left side and a right side, wherein the left sides of the plurality of extending blocks are aligned with a left face of the first and second plates and the right sides of the plurality of extending blocks are aligned with a right face of the first and second plates.
6. The device of claim 5, wherein the angled surface forms a continuous sheet between the left and right sides of the plurality extending blocks in order to increase the surface area of the angled surface.
7. The device of claim 1, further comprising a lock coupled with the positioning element and configured to physically bias the rotational orientation of the positioning element into one of a plurality of positions.
8. The device of claim 7, wherein the lock comprises one or more stoppers each having a bump and a dial having one or more dimples and coupled with the positioning element such that the dial rotates with the positioning element, wherein the bumps do not rotate with the dial and the stoppers are positioned adjacent to the dial such that, when aligned, one or more of the bumps spring into one or more of the dimples.
9. The device of claim 8, further comprising one or more support panels coupled with the lock and the extending mechanism, wherein each of the one or more support panels are positioned within a panel aperture on each of the plates such that as the plates move between the retracted and the

extended positions the plates slide up or down the panels via the panels apertures.

10. The device of claim 9, wherein at least one of the one or more support panels comprises a pair of grip tabs that protrude from the sides of the support panel into a pair of grip apertures formed by the plates when the plates are in the retracted position.

11. A bone fusion device for insertion into a desired location comprising: an extending mechanism including one or more extending blocks coupled with a positioning element such that movement of the positioning element causes the one or more extending blocks to move with respect to the positioning element; a first plate; a second plate, the first and second plates straddling the extending mechanism and coupled with the one or more extending blocks such that when the one or more extending blocks move with respect to the positioning element, the first and second plates move along a path with respect to each other between a retracted position in which the first and second plates are in contact with each other to an extended position in which the plates are spread apart from and do not contact each other; and a lock having a body that contacts the positioning element to physically bias the movement of the positioning element into one of a plurality of positions.

12. The device of claim 11, wherein when in the extended position, the first and second plates are spread apart from and do not directly contact each other.

13. The device of claim 11, further comprising one or more biasing elements coupled with the first plate and the second plate and positioned such that in the extended position the one or more biasing elements extend from the first plate to the second plate, wherein the one or more biasing elements have a shape selected from the group consisting of a ring, a C-shape and a ring-shaped coil.

14. The device of claim 11, wherein the first and second plates are sized such that the entirety of the perimeter of the first and second plates about the path align with the outermost perimeter of the device about the path.

15. The device of claim 11, wherein the bone fusion device further comprises one or more support panels coupled with the extending mechanism, wherein each of the one or more support panels are positioned within a panel aperture on each of the first and second plates such that as the first and second plates move between the retracted and the extended positions the first and second plates slide up or down the panels via the panels apertures.

16. The device of claim 15, wherein at least one of the one or more support panels comprises a pair of grip tabs that protrude from the sides of the support panel into a pair of grip apertures formed by the first and second plates when the first and second plates are in the retracted position.

17. The device of claim 11, wherein the angled surface forms a continuous sheet between the left and right sides of the one or more extending blocks in order to increase the surface area of the angled surface.

18. The device of claim 11, wherein the lock comprises one or more stoppers each having a bump and a dial having one or more dimples and coupled with the positioning element such that the dial rotates with the positioning element, wherein the bumps do not rotate with the dial and the stoppers are positioned adjacent to the dial such that, when aligned, one or more of the bumps spring into one or more of the dimples.

19. The device of claim 18, further comprising one or more support panels coupled with the lock and the extending mechanism, wherein each of the support panels are positioned within a panel aperture on each of the first and second plates such that as the first and second plates move between the retracted and the extended positions the first and second plates slide up or down the panels via the panels apertures.

20. The device of claim 19, wherein at least one of the one or more support panels comprises a pair of grip tabs that protrude from the sides of the support panel into a pair of grip apertures formed by the first and second plates when the first and second plates are in the retracted position.
